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The development of genetically improved red tilapia lines through the backcross breeding of two *Oreochromis niloticus* strains

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ABSTRACT

In order to develop a crossbred of *Oreochromis niloticus* with a high growth rate and coloration, a backcross breeding program was developed using Red-Stirling and Chitralada strains with 50%, 75%, and 87.5% of genic proportion to Chitralada. Seven genetic groups were obtained from Chitralada (C) and Red-Stirling (R): the hybrids F_1, fish-female Red-Stirling × fish-male Chitralada (RC), and its reciprocal (CR), and the backcross breeding (BB1), fish-female CR × fish-male Chitralada (CR × C), and its reciprocal (C × CR) and BB2 [C(C × CR)]. Growth curves were adjusted using an exponential model for females and males. The curves show a distinct difference among genetic groups and between genders based on the backcross breddingapproach. The best ranking for red females was observed in BB1 [C × (CR)]. The gain observed is due to the action of a genetic additive effect of Chitralada (42.90 g), a maternal effect of Chitralada (41.48 g), and a non-additive effect, which is attributed to paternal heterosis (11.41 g). While the best ranking for red males was found in BB2, The observed gain may be attributed to an additive genetic effect of Chitralada (74.04 g) and maternal effect of Chitralada (57.68 g). Survival among females was average. Among males, a higher mortality in the Red-Stirling group was observed compared to the other groups. These results indicate that the backcross breeding program can be used to develop genetic lines of red tilapia for further genetic breeding programs mainly exploring paternal heterosis to produce dams.
Statement of relevance: The present study proposes a breeding methodology based in robust statistical analysis to develop genetically lines of red tilapia.

1. Introduction

The per capita global consumption of fish has increased from 9.9 kg in 1960 to 19.2 kg in 2012 (FAO, 2014). China remains the world's largest producer of aquaculture fish and Brazil ranks 12th in aquaculture production (707.461 tons in 2012) (FAO, 2014). Aquaculture is the sector of food production with the fastest growth rate in Brazil, with an increase in 61.4% between 2009 and 2011. The Nile tilapia, *Oreochromis niloticus*, represents a species with the highest production (169,306 tons in 2013), corresponding to 43% of the total of inland aquaculture production (IBGE, 2013). Considering this recent growth and intensification in aquaculture production, there is a need for new technologies and higher performance strains to make a more efficient and sustainable supply chain.

Red tilapia was first discovered in Taiwan in the 1960s (Kuo and Tsay, 1984; Kuo, 1988). Since then, different strains of red tilapia have been commercially grown and many hybrids have been created from different species, predominantly *Oreochromis mossambicus* and *O. niloticus* (Koren et al., 1994; Romana-Eguia et al., 2004). Red-Stirling is a red variety of *O. niloticus* that produces all-red progeny due to a single autosomal dominant gene (McAndrew et al., 1988; Hussain, 1994). Red tilapia sales have grown in some markets due to a greater acceptance of red varieties, as well as the recognition of the value it adds to the final product (Josupeit, 2005; Hamzah et al., 2008; Thodesen et al., 2013).

Selection programs and interspecific crossbreeding have been developed in order to improve the production efficiency of red tilapia (Behrends et al., 1990; Thodesen et al., 2013). Methods based on backcross breeding have aimed to increase the genetic variety of higher performance strains and to maintain phenotypes of interest (which have not been reported in the literature), with regard to the development

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of red tilapia strains with high performance comparable to its wild-colored counterparts. The backcrossing method has been used in livestock breeding improvement programs to manage phenotypes controlled by a few genes (Hillel et al., 1993; Visscher and Haly, 1999; Giovambattista et al., 2000). Behrends and Smitherman (1984) developed what they refer to as "introgressive breeding techniques," or hybridization, in order to create a cold tolerance red tilapia strain crossing *Oreochromis aureus* with red tilapia of unknown origin. Behrends et al. (1990) also evaluated different approaches, such as two-stage hybridization, hybridization followed by repeated backcrossing, and bidirectional-backcross selection using *O. aureus* and an unknown red hybrid for developing red tilapia hybrids with superior performance. Tave et al. (1990a, 1990b) used reciprocal intraspecific backcrossing of two different strains of *O. niloticus* to produce cold-tolerant and high performance strains of Nile tilapia.

In this study, a breeding improvement program was developed to evaluate the use of backcross in two Nile tilapia strains (Chitralada and Red-Stirling) in order to test the technique to develop synthetic strains of red tilapia for a future breeding program. The performance and effects of maternal and paternal individual heterosis on groups with different genetic proportions of Chitralada obtained by backcross breeding between Chitralada and Red-Stirling were evaluated.

2. Materials and methods

2.1. Breeding program location

The breeding program was carried out in the Brazilian Industry of Fish Ltd. Hatchery facilities, which is located in Itupeva city (23°20′S; 47°02′W) in the State of São Paulo, Brazil, the Itupeva region characterized by a humid subtropical climate (Cwa).

2.2. Genetic strains and crossbreeding

The Red-Stirling originating from University of Stirling as fry in 2000 was established as broodstock after 10 years of mass section for coloration and growth-rate. The Chitralada strain was acquired in 2000 from a governmental hatchery and was mass-selected for growth-rate. Genetic groups were produced from a mating system that consisted of with one male for every two females, with 30 males and 60 females used in each one of the seven genetic groups. The Generation F_1 was obtained by breeding Red-Stirling and Chitralada. Subsequently, two generations of backcross breeding (BB1 and BB2) were carried out with Chitralada. No selection was performed on Red-Stirling and Chitralada fish before crossing. These crosses were repeated using the stock groups in Brazilian Industry of Fish Ltd. Thus, getting contemporary genetic groups for this study. Seven reproduction hapas ($8 \times 6 \times 2$ m) in a land farm were set in diallel (Table 1), totaling seven genetic groups with 50% (F1), 75% (BB1), and 87.5% (BB2) of Chitralada genetic proportions.

The broodstock were fed to apparent satiation with an appropriate commercial feed adequate to each stage of the culture three times a day (42% crude protein for the restock stage and 32% for the growth phase). Water quality parameters were measured. Temperature and dissolved oxygen were measured daily using a digital oximeter (YSI, Model 55 Hexis) and ammonia and pH were measured weekly using a commercial kit.

Table 1
Diallel crosses between Red-Stirling and Chitralada strains.

Female	Male			
	Red Stirling (R)	Chitralada (C)	F_1 (CR)	RC1 (C × CR)
Red-Stirling (R)	Red-Stirling (R)	F_1 (RC)	–	–
Chitralada (C)	F_1 (CR)	Chitralada (C)	BB1(C × CR)	BB2 [C(C × CR)]
F_1 (CR)	–	BB1 (CR × C)	–	–

Hapas were daily inspected and larvae (fry) were weekly collected across four weeks. Larvae were divided into $1 \times 1 \times 1$ m hapas mounted in a greenhouse. Once they achieving a minimum weight of 15 g, fingerlings were mass-select by color. Wild individuals were eliminated, except for the group Chitralada. Those selected were sexed, and marked with electronic microchips (tagged with a Passive Integrated Transponder (PIT) tags). Fifteen dams and fifteen sires from each hapa were randomly selected and classified by sex and into four classes based on weight. For males the class limits were: 1 = 18–28 g; 2 = 28–32 g; 3 = 32–38 g and 4 = 38–50 g, with 105 fishes each class. For females were: 1 = 12–22 g; 2 = 22–26 g; 3 = 26–32 g and 4 = 32–50 g, with 105 fish in each class. A total of 420 females and 420 males were used.

2.3. Grow-out performance

Tagged fingerlings were placed into eight net cages, four for females and four for males. Each fingerling class was raised in one net cage, having each cage all classes. The statistical design chosen for the grow-out evaluation was completely randomized, in which each animal was an experimental unit and the model was corrected using the covariate of age. Every 21 days during 190 days of grow-out, the fish were anesthetized with benzocaine (0.01%) to obtain weight and standard length. Weight was measured using a balance with a 1 g precision and the standard length was measured with the aid of an analog caliper. After biometrics were taken, the fish were returned to their respective hapas. Morphometric measurements were taken every 21 days across 190 days of cultivation.

2.4. Statistical analysis

Statistical analyses were performed by R software version 3.1.2 (R Development Core Team, 2012) using *lmer ()* and *ranef ()* functions from the *lme4* library (Bates et al., 2012) to estimate and predict the mixed model (using the restricted maximum likelihood method). Thus, these analyses enabled group rankings according to breeding values (BLUP) to define the best groups. A main mixed model was used for all analysis, which each fish was regarded as an experimental unit. The model can be written as:

$$Y_{jimkl} = m + c_l + b_k + p_m + g_i + a_j + e_{ijkl};$$
$$\text{with}: e_{jimkl} \sim (0, \sigma_e^2); g_i \sim \left(0, \sigma_g^2\right); a_j \sim (0, \sigma_a^2);$$

where Y_{jimkl} is the observation of the fish response variable j belonging to the genetic group i, of the week of birth k and net cage j; m is a constant inherent to all observations; c_l is the effect of l-th net cage ($l = 1, 2, 3, 4$); b_k is the effect of k-th week of birth ($k = 1, 2, 3, 4$); p_m is the effect of m-th evaluation ($m = 1, 2, 3, 4$); g_i is the effect of i-th genetic group ($i = 1, 2, 3, \ldots, 7$) (random effect); a_j is the effect of the j-th fish (j = 1, 2, …, 622); e_{jimkl} is a random, unobservable variable.

To evaluate weight gain across different genetic groups, nonlinear regression models were adjusted to body weight data depending on age. Growth models Gompertz = Aexp(-Be^{-Kt}), logistic = $A(1 + Be^{-Kt})^{-1}$, and exponential = Ae^{Kx} were tested. Due to the heterogeneity of variances in body weight, a model weighted by the inverse of the variance of weights was adopted. The selection of the best model in the first phase was made using the Akaike information criterion (AIC) and Bayes Information Criterion (BIC). The lower the values of AIC and BIC, the better the model.

To adjust non-linear regression models, the functions *nlsList ()* and *nlme ()* from *nlme* library were used (Pinheiro et al., 2015). Through a graphical analysis of residues, the presence of heteroscedasticity in all adjusted models was found. To correct for this problem, the argument *weights var* = *Const Power* of the *nlme ()* function was used.

Genetic components for body weight were estimated using the least squares method (Dickerson, 1973) across genetic groups in both males and females. Were used the test of deviation to check the significance of the data adjustment. The additive genetic effect of Red-Stirling (g1), the additive genetic effect of Chitralada (g2), the maternal effect of Red-Stirling (m1), the maternal effect of Chitralada (m2), the individual heterosis (hi), the maternal heterosis (hm), and the paternal heterosis (hp) were measured. Paternal and maternal heterosis is regarded as partitions of the heterosis effects (Maluwa and Gjerde, 2006).

For survival analysis, a generalized linear mixed model with link function *logit* was used with binomial values related to mortality. The data were submitted to model adjustment through the function *glmer ()* from the *lme4* library (Bates et al., 2012). To verify the group effect, an analysis of deviance was made through *Anova ()* function from the *car* library (Fox and Weisberg, 2011). The group effect was verified by analysis of deviance and, in the case of a significant effect, the adjusted means and confidence intervals were obtained by the *LSmeans ()* function from the *doby* library (Højsgaard and Halekoh, 2013).

3. Results

3.1. Test environment

The following environmental parameters were obtained during the experimental grow-out period: average water temperature of 23.5 °C, dissolved oxygen of 5.2 mg L^{-1}, pH of 7.0, and ammonium lower than 0.06, on average. All the parameters were within the requirements of water quality for tilapia cultivation (Balarin and Hatton, 1979).

3.2. Grow-out performance

The exponential model showed the best adjustment for body weight according to age. There was a convergence of Gompertz and Logistic models; the adjustment was obtained for growth separately for males and females in each genetic group. The parameters "A" and "k" from the weight model were estimated according to age, in which "A" represents the estimate of the initial weight and "k" refers to the specific growth rate (Table 2). These parameters were significantly different across genetic groups in both males and females. The Red-Stirling group had lower weight gain, ranked last place, and had a lower specific growth rate (0.014 and 0.016) in females and males, respectively. The crosses demonstrated, in general, intermediate values that were higher than the parental Red-Stirlings; however, these values were lower than the parental Chitralada in both genders.

Growth curves were adjusted using an exponential model for females (Fig. 1) and males (Fig. 2) in different genetic groups; the curves show a distinct difference among genetic groups and between genders. The average values (corrected by the model), genetic values and

Table 2
Parameter estimates (A and K) of exponential growth model for body weight according to age for females and males of genetic groups.

Genetic groups	Female		Male	
	A	K	A	K
Parental				
Red-Stirling	5.19 b	0.014 b	4.25 b	0.016 b
Chitralada	2.85 a	0.019 a	2.13 a	0.023 a
F_1				
RC	3.30 ab	0.018 a	1.85 a	0.025 a
CR	4.52 ab	0.016 ab	3.65 b	0.019 b
Backcross breeding				
BB1 (CR) × C	4.03 ab	0.017 ab	2.54 ab	0.021 ab
BB1 C × (CR)	4.95 b	0.017 ab	2.47 ab	0.021 ab
BB2	3.09 a	0.019 a	3.26 ab	0.020 ab

Estimates followed by different letters in the same column differ by not overlapping confidence intervals at 95% probability.

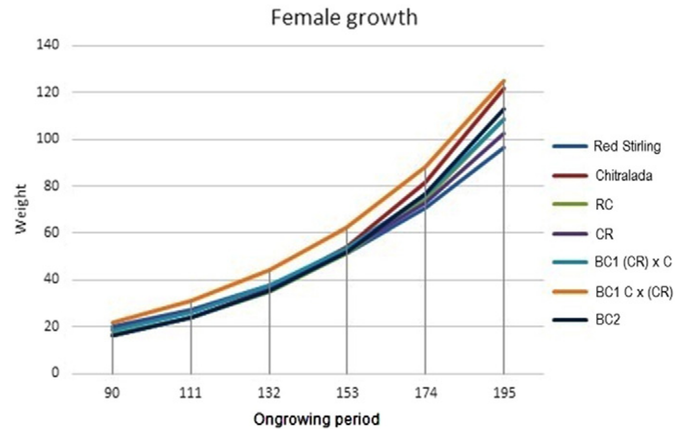

Fig. 1. Growth curves adjusted by the exponential model for females of different genetic groups during the 190 days of grow-out testing.

standard length at 190 days of cultivation were obtained (Table 3). For females, the BB1 [C × (CR)] group had the highest gain (breeding value of 3.49), even higher than the parental Chitralada. While for males, the best group was the Chitralada (breeding value of 9.05) followed by the BB2 group (1.42). The Red Stirling group was the worst in both genders.

3.3. Estimates of genetic components of body weight at harvest and survival

The additive effects of Red-Stirling strain (g1), the additive effects of Chitralada strain (g2), the maternal effects of Red-Stirling strain (m1), the maternal effects of Chitralada strain (m^2), the individual heterosis (h), the maternal heterosis (hm), the paternal heterosis (hp) for body weight of females and males during the grow-out period (Table 4; Table 5).

The gain in F1 groups can be attributed to heterosis (7.18 g for males and 4.43 g for females). The gain observed in BB1 [C × (CR)] for females is due to the action of a genetic additive effect of Chitralada (42.90 g), a maternal effect of Chitralada (41.48 g), and a non-additive effect, which is attributed to paternal heterosis (11.41 g). In males, paternal heterosis was negative for BB1 [C × (CR)] and BB2 groups. The observed gain may be attributed to an additive genetic effect of Chitralada (74.04 g) and maternal effect of Chitralada (57.68 g).

No significant differences ($P > 0.05$) for survival were detected among the genetic groups for females, where the overall survival was 80%. Conversely, significant differences were observed for males ($P < 0.05$). The Red Stirling group showed the lowest survival (66%).

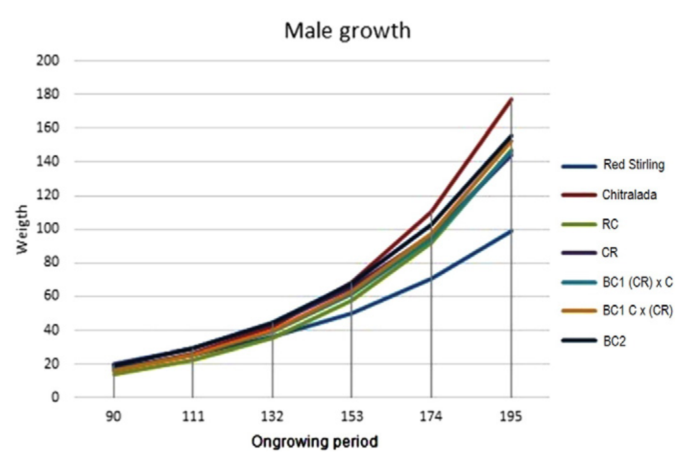

Fig. 2. Growth curves adjusted by the exponential model for males of different genetic groups during the 190 days of grow-out testing.

Table 3
Adjusted means of weight and standard length of male and female progeny crosses with 190 grow-out days and their genetic values and ranking.

	Female				Male			
	Weight (g)	Length (cm)	Breeding value	Ranking	Weight (g)	Length (cm)	Breeding value	Ranking
Parental								
Red-Stirling	73.73	12.93	−7.84	7°	86.13	13.57	−10.98	7°
Chitralada	101.89	14.40	2.38	2°	143.67	16.04	9.05	1°
F_1								
RC	92.25	13.99	0.54	5°	120.82	15.03	1.32	3°
CR	89.01	14.04	−1.74	6°	121.96	15.38	−0.60	6°
Backcross breeding								
BB1 (CR) × C	93.44	14.05	0.88	4°	121.48	15.16	−0.51	5°
BB1 C × (CR)	108.47	14.88	3.49	1°	125.73	15.28	0.28	4°
BB2	95.54	14.28	2.30	3°	130.58	15.65	1.42	2°

According to the confidence interval the other groups did not differ. The confidence interval across groups, as well as the survival values, can be found in Fig. 3.

4. Discussion

4.1. Grow-out performance

In this study, there were differences in growth patterns between males and females. Over the course of approximately 140 days, growth for males and females was similar, as shown in Figs. 1 and 2. Oliveira et al. (2013) found that the growth of males and females of Nile tilapia GIFT had similar weights until 160 days of age. After this period, growth patterns among groups for both males and females were differentiated in growth and there was a change in group rankings according to gender. For females the best group was the BB1 C × (CR) that presented breeding value of 3.49, being the first in the ranking. While for males, the best group was the Chitralada first in the ranking with breeding value 9.05, followed by the BB2 group with 1.42.

The Red-Stirling group had lower weights, ranked last place, and had a lower specific growth rate (0.014 and 0.016) in females and males, respectively, although they had a higher initial estimated weight. The performance of the Red-Stirling group was weaker compared to other varieties, such as the Chitralada, Malaysia, and Taiwan and Thailand red *Oreochromis* spp. (Moreira et al., 2005; Nguyen et al., 2009). These studies corroborate the results found in the present evaluation. There are no records of genetic improvement programs to improve growth or weight gain in the Red-Stirling variety, which could explain its lower performance compared to higher performing strains or crossbreeding strains.

The F_1 groups demonstrated intermediate values that were higher than the parental Red-Stirlings; however, these values were lower than the parental Chitralada in both genders. Moreira et al. (2005) studied crossbreeding between Chitralada and Red-Stirling and did not find any significant differences in growth parameters among F_1 hybrids and Chitralada. Crossbreeding using Chitralada females produced higher gains compared to the F_1 groups. The BB1 [C × (CR)] group had the highest gain, even higher than the parental Chitralada, suggesting that they were affected by non-additive genetic components. In males, the Chitralada group had the greatest weight, followed by BB2; both groups had the highest specific growth rate. The growth pattern of males and females is different in tilapia due to a sexual dimorphism (Oliveira et al., 2013; Lind et al., 2015). Studies on sexual dimorphism in species and strains of tilapia suggest that strategies should be tailored specifically for males and females in order to achieve optimal results (Lind et al., 2015). The difference in ranking among groups is possibly due to the distinct genetic components, such as additive and non-additive genetic effects of males and females heterosis.

4.2. Genetic parameters of crossbreds

F_1 groups had greater weight increases compared to parental Red-Stirling. This gain can be attributed to heterosis (7.18 g for males and 4.43 g for females). Although the observed heterosis has been positive, low values were demonstrated. Studies involving four strains of *Oreochromis* spp., red tilapia from Malaysia, Stirling, Taiwan, and Thailand in a complete diallel cross, indicated that the average heterosis for body weight was low (approximately 4.2%) and no significant differences were observed between the average weight of crossings and parental averages (Nguyen et al., 2009). The non-additive genetic

Table 4
Genetic components for body weight of the different genetic groups for females. G1 is the additive genetic effect of the Red-Stirling strain, g2 is the additive genetic effect of the Chitralada strain, m1 is the maternal effect of the Red-Stirling strain, m2 is the maternal effect of the Chitralada strain, hi is the individual heterosis, hm is the maternal heterosis, and hp is the paternal heterosis.[a]

	Genetic components (g) for female						
	g1	g2	m1	m2	h	hm	hp
Parental							
Red-Stirling	29.01	0	44.72	0	0	0	0
Chitralada	0	57.20	0	41.48	0	0	0
F_1							
RC	14.51	28.60	44.72	0	4.43	0	0
CR	14.51	28.60	0	41.48	4.43	0	0
Backcross breeding							
BB1 (CR) × C	7.25	42.90	22.36	20.74	2.22	2.02	0
BB1 C × (CR)	7.25	42.90	0	41.48	2.22	0	11.41
BB2	3.63	50.05	0	41.48	1.11	0	5.71

[a] The model deviations were not significant it indicates that the data set to the model.

Table 5
Genetic components for body weight of the different genetic groups for males. G1 is the additive genetic effect of the Red-Stirling strain, g2 is the additive genetic effect of the Chitralada strain, m1 is the maternal effect of the Red-Stirling strain, m2 is the maternal effect of the Chitralada strain, hi is the individual heterosis, hm is the maternal heterosis, and hp is the paternal heterosis.[a]

	Genetic components (g) for male						
	g1	g2	m1	m2	hi	hm	hp
Parental							
Red-Stirling	29.59	0	56.54	0	0	0	0
Chitralada	0	84.62	0	57.68	0	0	0
F_1							
RC	14.80	42.31	56.54	0	7.18	0	0
CR	14.80	42.31	0	57.68	7.18	0	0
Backcross breeding							
BB1 (CR) × C	7.40	63.47	28.27	28.84	3.59	10.08	0
BB1 C × (CR)	7.40	63.47	0	57.68	3.59	0	7.77
BB2	3.70	74.04	0	57.68	1.80	0	3.89

[a] The model deviations were not significant it indicates that the data set to the model.

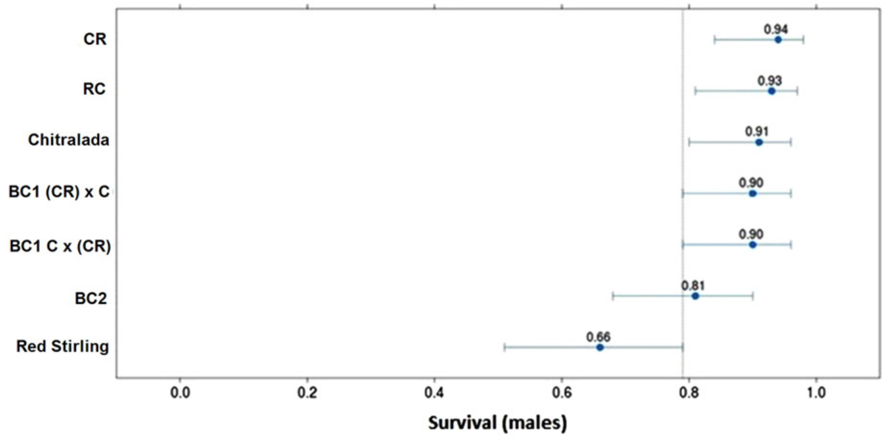

Fig. 3. Confidence interval between the observed survival of groups and values for males.

components (the average level of heterosis for body weight in fish) are generally low. The comparison between the studies may not be accurate since the expression of heterosis depends on the combinations among varieties (Pongthana et al., 2010), although dominance effects mainly explain heterosis. In general, crossbreeding organisms from genetically distinct populations or between inbred lines should result in higher heterosis from the crossing than from nearby populations (Falconer and Mackay, 1996; Pongthana et al., 2010).

The gain observed in BB1 [C × (CR)] for females is due to the action of a genetic additive effect of Chitralada, a maternal effect of Chitralada, and a non-additive effect, which is attributed to paternal heterosis (11.41 g). Thus, in females, BB1 [C × (CR)] stands out as the first in the ranking to present higher values of both weight and length compared to parental Chitralada. Tave et al. (1990b) attributes the gains observed for weight and standard length of F_2 generations and backcrossing among the varieties of Egyptian *O. niloticus* and Ivory Coast to maternal heterosis; however, these authors investigated the effects of paternal heterosis on the observed results. In males, paternal heterosis was negative for BB1 [C × (CR)] and BB2 groups; however, the purpose of the study, gene introgression, was achieved with the BB2 group, which occupied the second position in ranking and had the best performance among the red-colored group. The observed gain may be attributed to an additive genetic effect of Chitralada (74.04 g), very close to the values obtained by pure Chitralada (84.62). Suggesting that backcross breeding was indeed effective for males in this program. Since there are no reports in the literature on the different effects of non-additive genetic components between males and females, additional studies are needed in order to understand the effects of heterosis across genders and how it manifests biologically.

4.3. Survival

The average survival among females was 80%, which is in accordance to Popma and Lovshin (1996). Among males, a higher mortality in the Red-Stirling group was observed compared to the other groups. The low rate of survival in the Red-Stirling tilapia can be explained by the aggressiveness of Chitralada males. Chitralada comes from an old genetic improvement program and has undergone many generations of selection. Schjolden and Winberg (2007), studying stress responsiveness in rainbow trout, reported that an intense selection processes can change the behavior and increase the levels of aggression and social stress.

Tilapia are often organized hierarchically. There is a dominant individual, usually a male, with priority access to resources, such as food and territory. Thus, a dominant emerges when there are variations in the environment or the group hierarchy is reset by confrontations and aggressive interactions that can cause death (Chellappa et al., 2012; Conrad et al., 2011). Therefore, it is possible that, when fish are classified according to weight and gender and allocated in cages, the hierarchy is reset, which favors Chitralada due to the greater proportion of the Chitralada gene present.

4.4. Developing synthetic strains of tilapia

Over the last decade, attempts to produce red tilapia strains with superior performance have been limited, possible due to the worldwide dominance of GIFT strains in tilapia production (Ponzoni et al., 2010). Thodesen et al. (2013) drew attention to the potential of the development of red tilapia with high salinity tolerance, which can be raised in brackish water, for the Chinese market. The authors worked on the Progift red tilapia strain, which had been developed from different Asia and Latin America tilapia genetic material. After five generations of full-sib family selection, this method resulted in 59% larger body weight at harvest with an improvement in blotched-free reddish tilapia.

The main goal of a red tilapia-breeding program is to combine the reduction of black blotching pattern with a growth rate comparable to that achieved for instance GIFT wild type tilapia. Mather (2001) applied mass selection in progenies from *O. niloticus* × *O. mossambicus* red hybrids and showed that the red phenotype can be improved without affecting growth rate. Halstrom (1984) also reported a reduction in black blotching on red tilapia through selective breeding; this reduction in black marking was also achieved after five generations of blotched-free mass selection of Red-Stirling tilapia (Garduño-Lugo et al., 2004).

The ranking of the genetic groups was different in relation to sex. The best weight performance for females was observed in BB1 [C × (CR)] and the best weight performance for males was observed in BB2. Red dams can be selected from the second generation of backcross breeding and sires can be selected from the third generation. These crossbred broodstock can be mass-selected during at least five generations to get a homozygous blotched-free red phenotype (Garduño-Lugo et al., 2004; Mather, 2001). From the outcomes of the backcross breeding used in the current study it is clear that the dominant red phenotype of the Red-Stirling strain can then be obtained by mass-selection to produce crossbred fish using different wild type strain. Red fish with different genetic backgrounds would then be the base of the population in order to establish a complete diallel-crossing breeding program and produce red tilapia with superior performance.

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